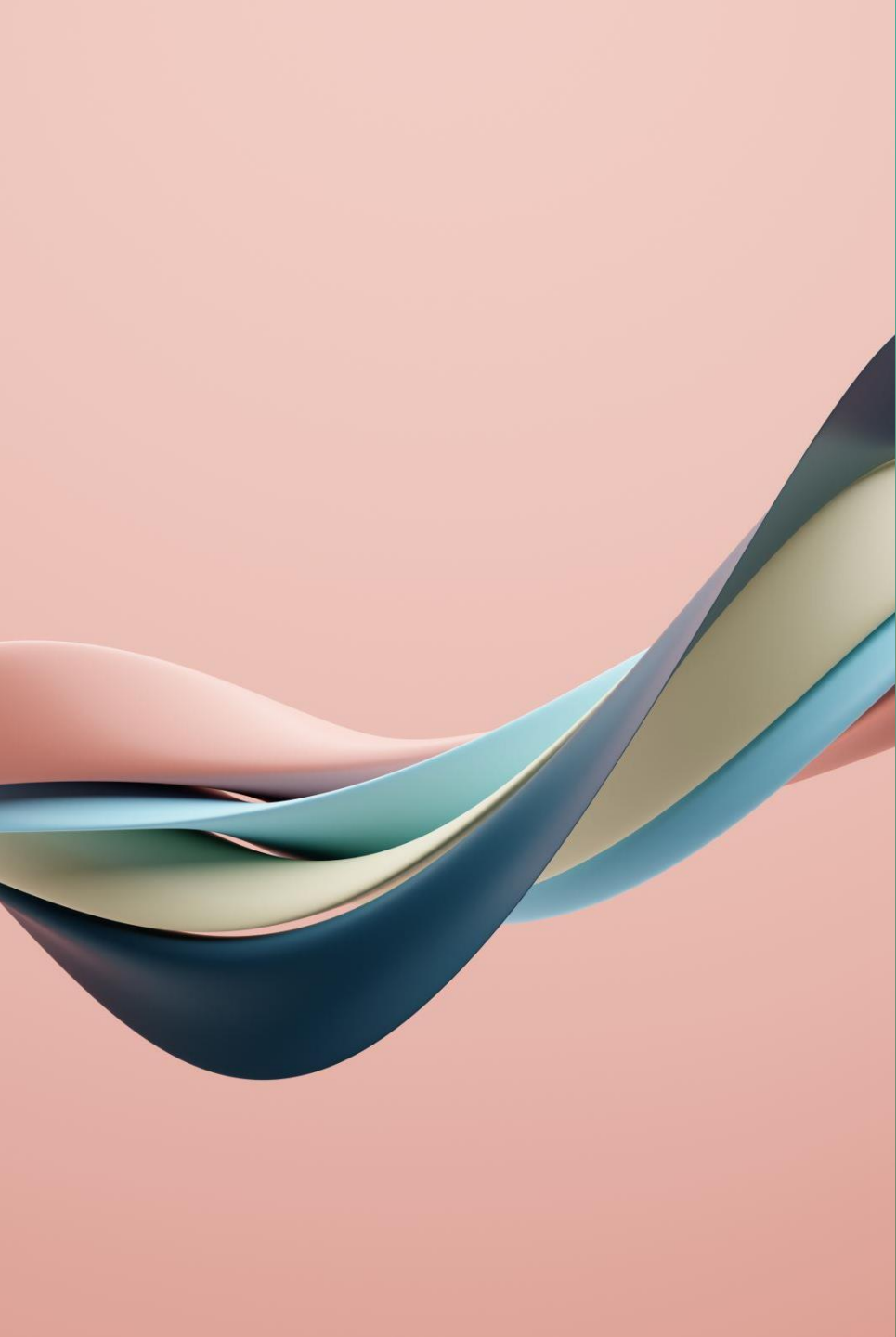




HCI-DESIGN AND EVALUATION (HCIDE) 2025

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XIANGHUA (SHARON) DING

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CLASS TIMETABLE

- Weekly lectures (JMS 438AB)
 - Friday: 1pm-3pm
- Weekly 1-hour labs (BOYD ORR: 1028 LAB)
 - Friday: 9am, 10am, 11am and 12pm
 - Lab starts next week
 - You will be automatically assigned to a group of 5-6 based on the lab you enrolled in



COURSE MATERIALS

- Moodle section for each week
 - Required reading
 - Lecture notes (these slides, available a day before the lecture)
 - Lab sheets (available at the start of the week)
 - Recorded lecture videos (do not rely on it)
 - Submission point for assessed exercises



HUMAN- COMPUTER INTERACTION (HCI)

Programing

Algorithms and
Data Structure

Database Systems

Network and
Operating Systems

Artificial
Intelligence

Information
Retrieval

Human-Computer
Interaction Design
and Evaluations

This course requires a different kind of effort than many other computing courses



WHY SHOULD CS CARE ABOUT HUMANS?
GO TO [MENTI.COM](https://www.menti.com) AND USE THE CODE



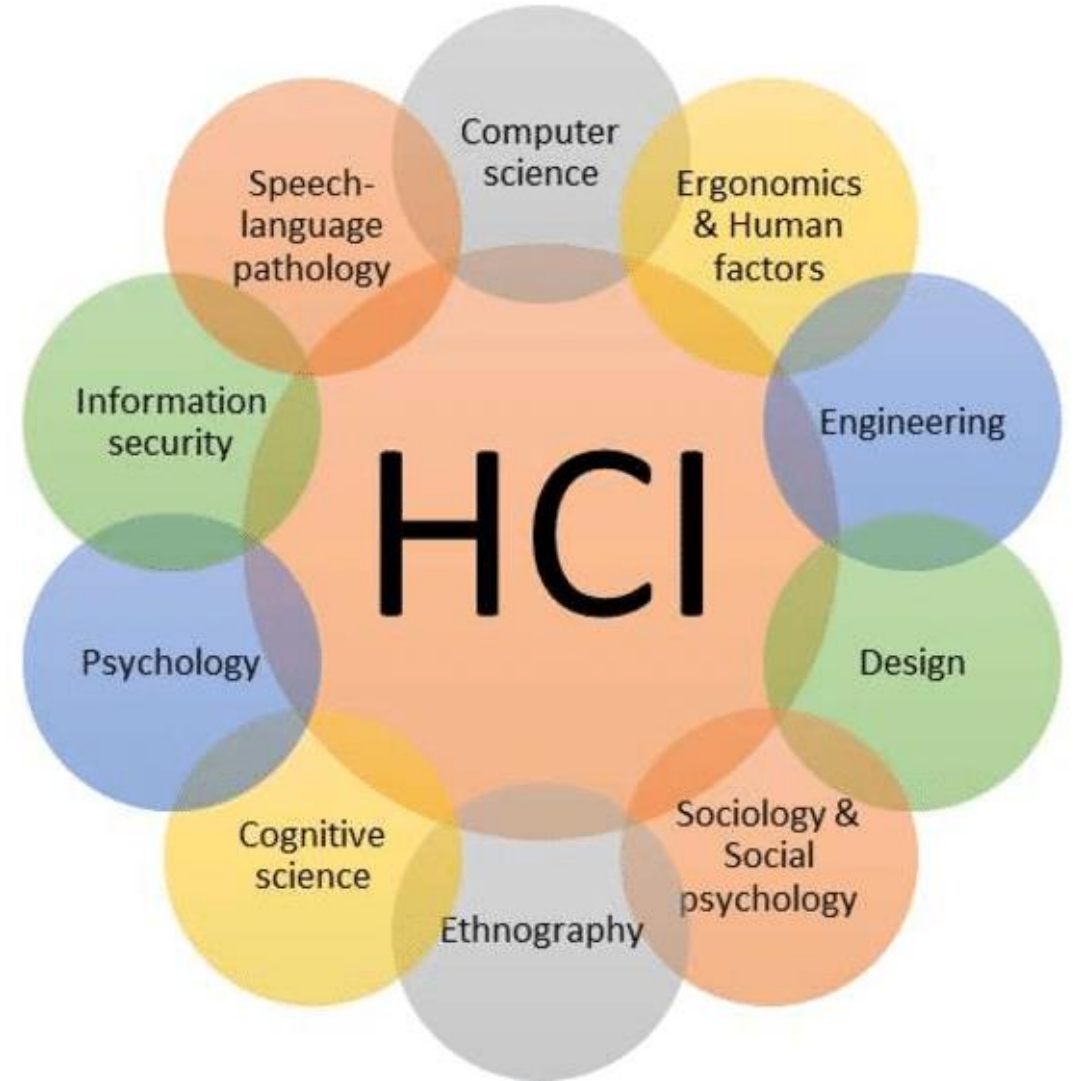
HUMAN-CENTERED COMPUTING?

HCI

The goal of this field is to solve real problems in the **design** and **use** of technology, making computer-based systems easier to use and more effective **for people and organizations**.



HCI is an **interdisciplinary** field.





SURVEY, OBSERVATIONS
AND INTERVIEWS



DESIGN IDEA
GENERATIONS



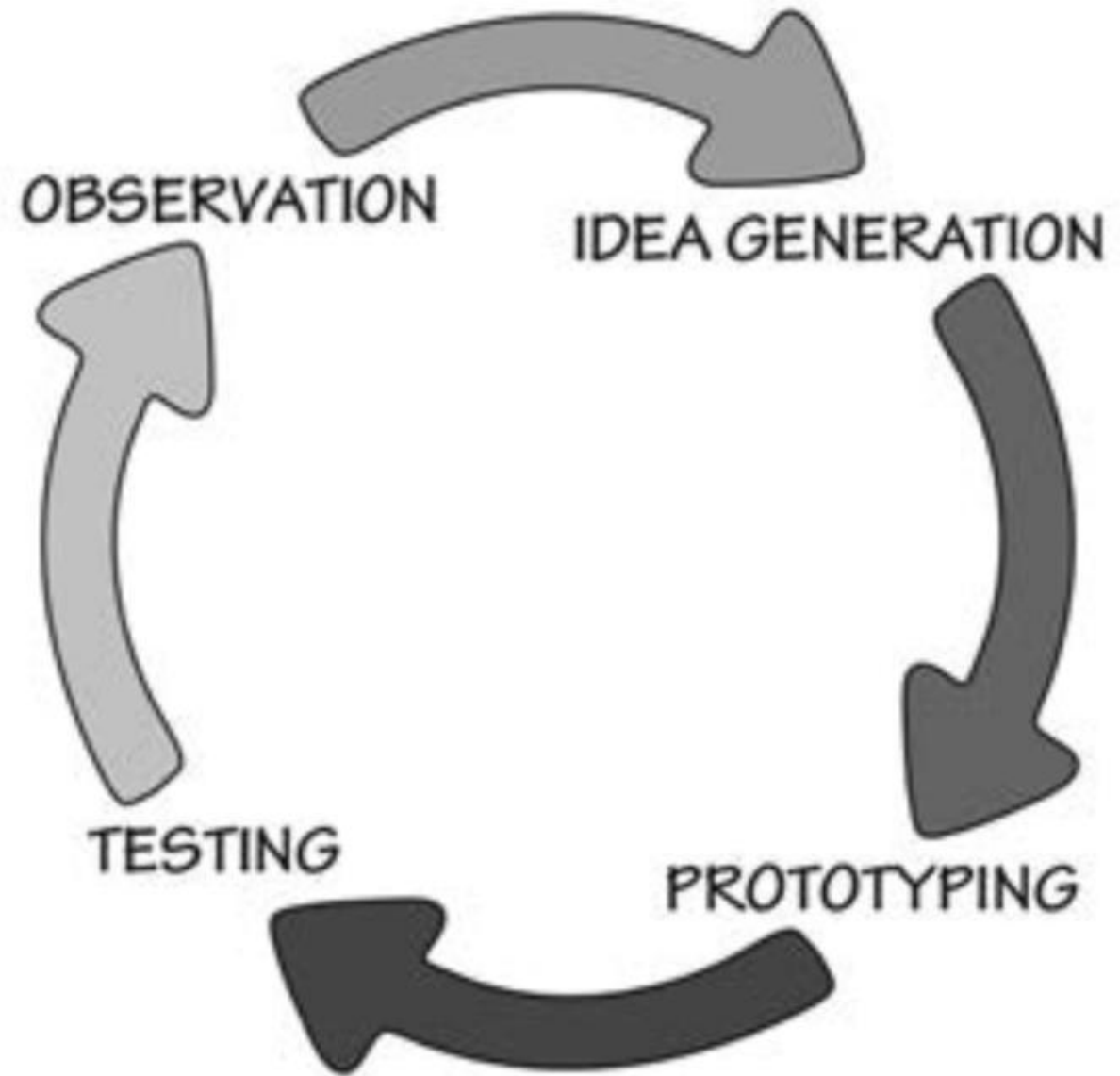
BUILDING PROTOTYPES



TESTING WITH HUMAN
PARTICIPANTS

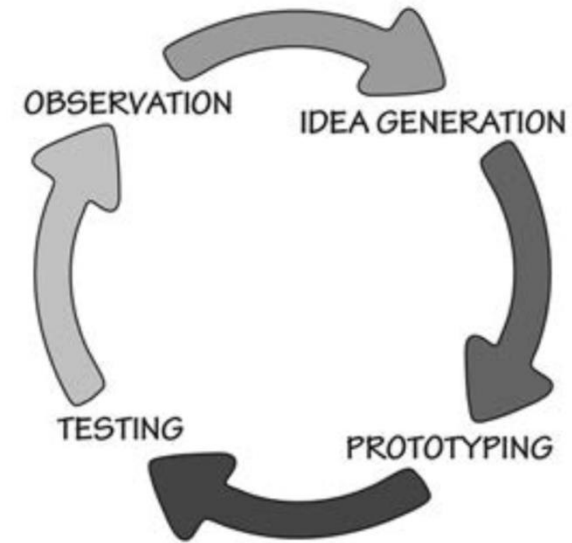
**WHAT ACTIVITIES ARE INVOLVED IN
HCI?**

HCI DESIGN CYCLE



HUMAN-CENTERED DESIGN

- What is the phenomenon/problem?
 - What is your idea?
 - Why do you think it will work?
 - Where is the proof?
-
- Collecting data from potential users
 - Can provide insights, but must be collected in a valid and theoretically sound way



**BY THE END OF THIS
COURSE, YOU WILL
KNOW HOW TO DO
THESE ACTIVITIES**

THE HISTORY OF HCI



DID HCI EXIST AT THE BEGINNING OF COMPUTING?
GO TO [MENTI.COM](https://www.menti.com) AND USE THE CODE

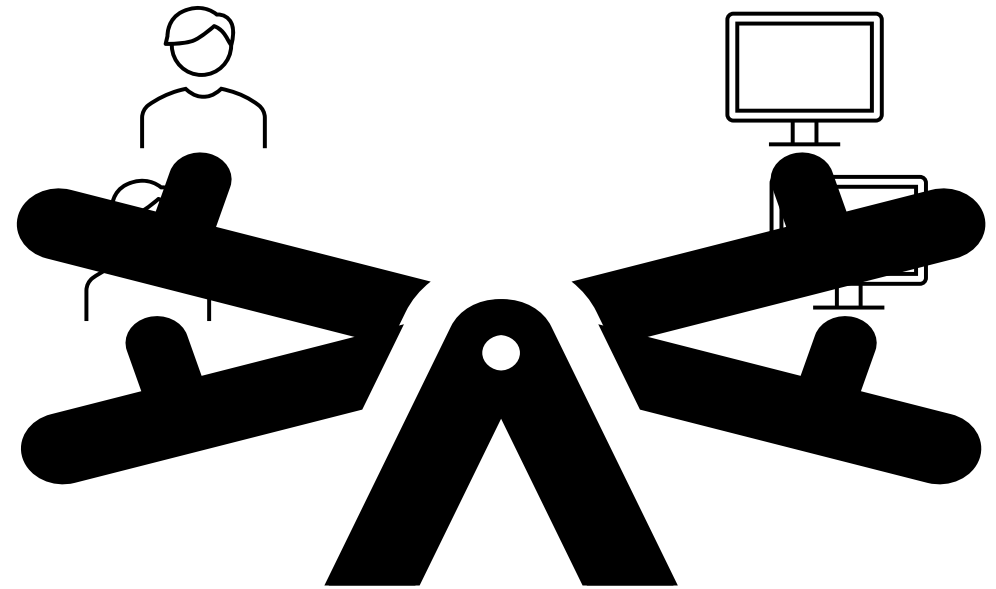
EARLY COMPUTERS

- The history of HCI is tied to the history of Computing
- Initial computers in research labs, took up full rooms
 - Only ever operated on by specialists/engineers/the people who built them



COMPUTERS BECOME AFFORDABLE

- As technology progressed, got smaller/more affordable, started appearing in workplaces and homes

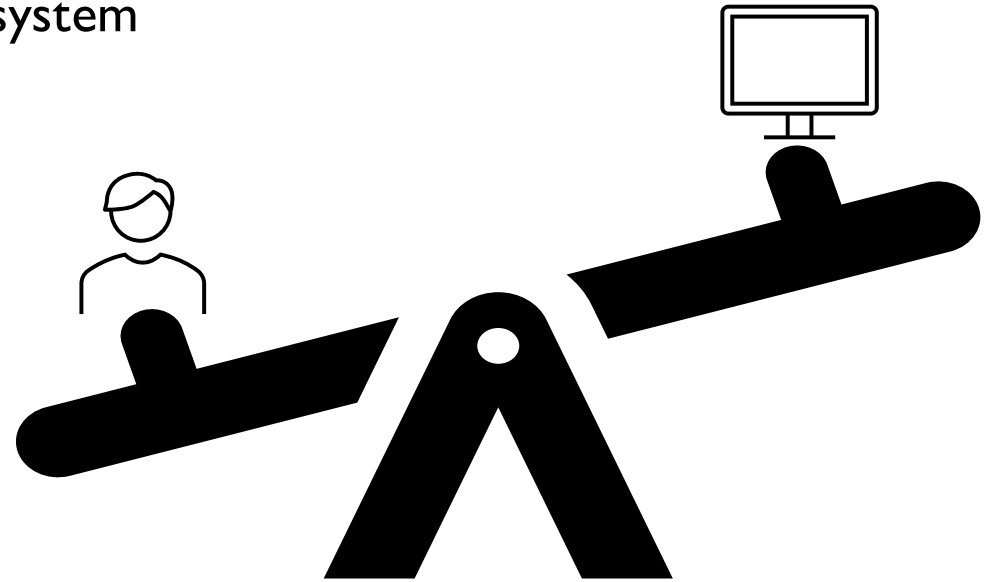




WHEN COMPUTING IS AFFORDABLE, SHALL
WE TRAIN PEOPLE—OR REDESIGN
COMPUTING?
GO TO [MENTI.COM](https://www.menti.com) AND USE THE CODE

COMPUTERS BECOME AFFORDABLE

- As technology progressed, got smaller/more affordable, started appearing in workplaces and homes
 - A need for 'real' people to be able to operate the system
 - A need for a new discipline to study these issues



HCI BEGAN

- HCI generally thought of as beginning in early 1980s
 - (Although studies done before, that in retrospect followed 'HCI' methods and principles)
 - Conferences began
 - Influential textbooks
 - Emergence of the Graphical User Interface

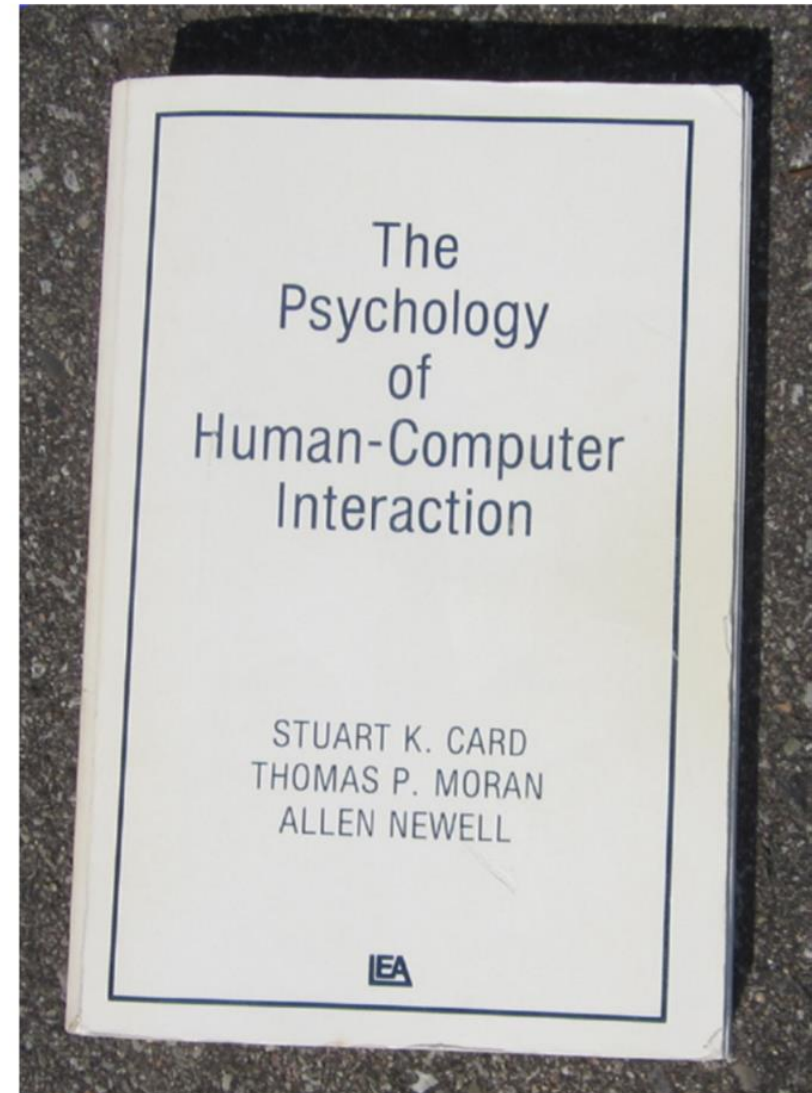
THE CONFERENCE

- 1983, the first ACM Human Factors in Computing Systems Conference, CHI



THE BOOK

- Trying to create an applied psychology of HCI
- Based on knowledge from human psychology
 - Perception
 - Cognition
 - Motor function
- For many in software design communities, first exposure to Psychology basics
- Engineering-style theories to give approx calculations of how efficiently humans would interact



GRAPHICAL USER INTERFACES

- Xerox Star – 1981
 - The first GUI computer released
 - WYSIWYG
 - Desktop metaphor
 - Yet not a commercial success
 - Very expensive; network terminal, not 'personal' computer
- Apple Macintosh – 1984
 - Brought the GUI to a wider audience

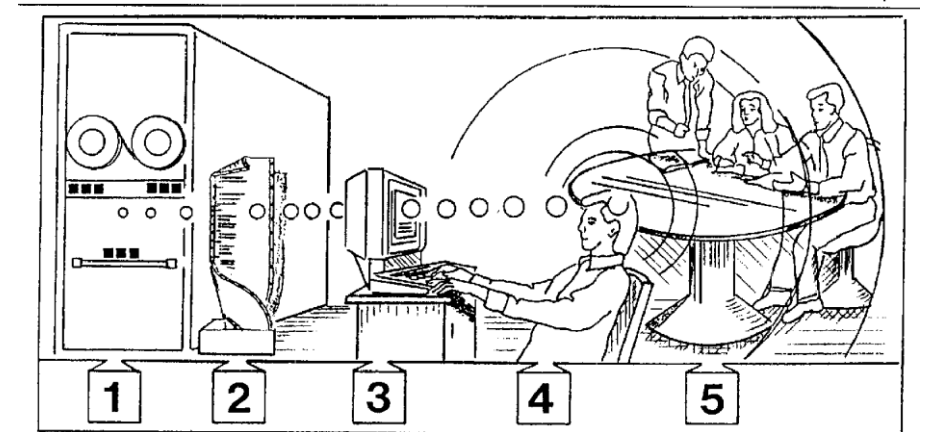


BROADENING OF HCI TOPICS

- ~1980s: early research often looking at usability and efficiency
 - e.g measure speed and accuracy
 - Lab-based studies
 - Formal experiments

BROADENING OF HCI TOPICS

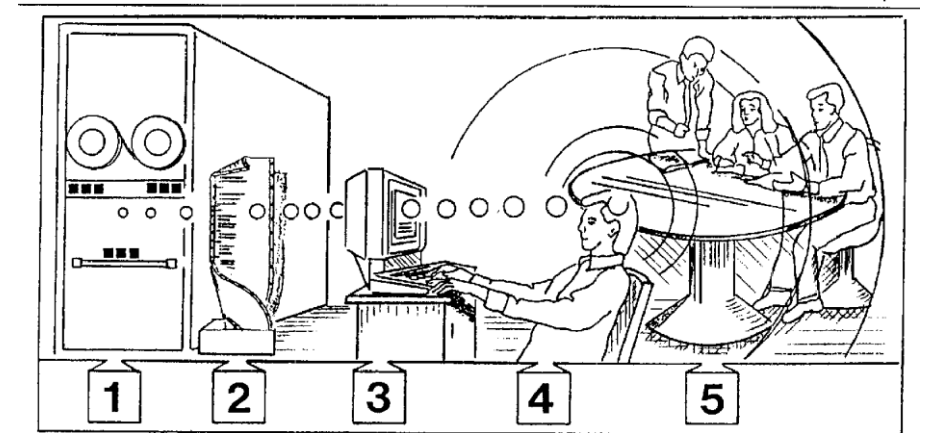
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 - Emails, Web: topics related to communication



“Computer Reaches Out”, Johnathan Grudin, 1991

BROADENING OF HCI TOPICS

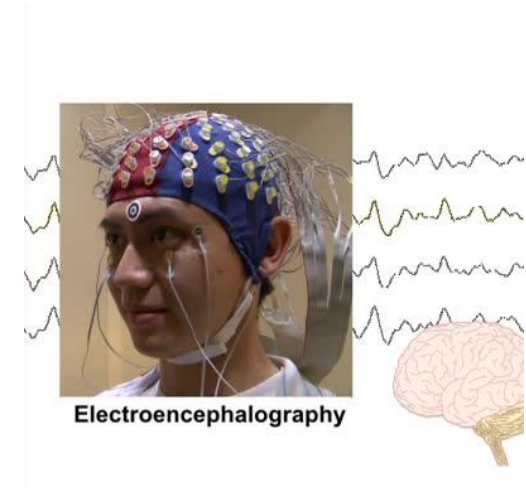
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 - e.g measure speed and accuracy
 - Lab-based studies
 - Formal experiments
- ~1990s: field started to broaden, alongside importance of Internet
 - Emails, Web: topics related to communication
- ~2000s: mobile/portable computing
 - Real world studies ‘in the wild’
 - New technologies: sensors, wearable, VR/AR...
 - Study social, emotional, cultural issues
 - “Older” forms of research haven’t gone away!



“Computer Reaches Out”, Johnathan Grudin, 1991

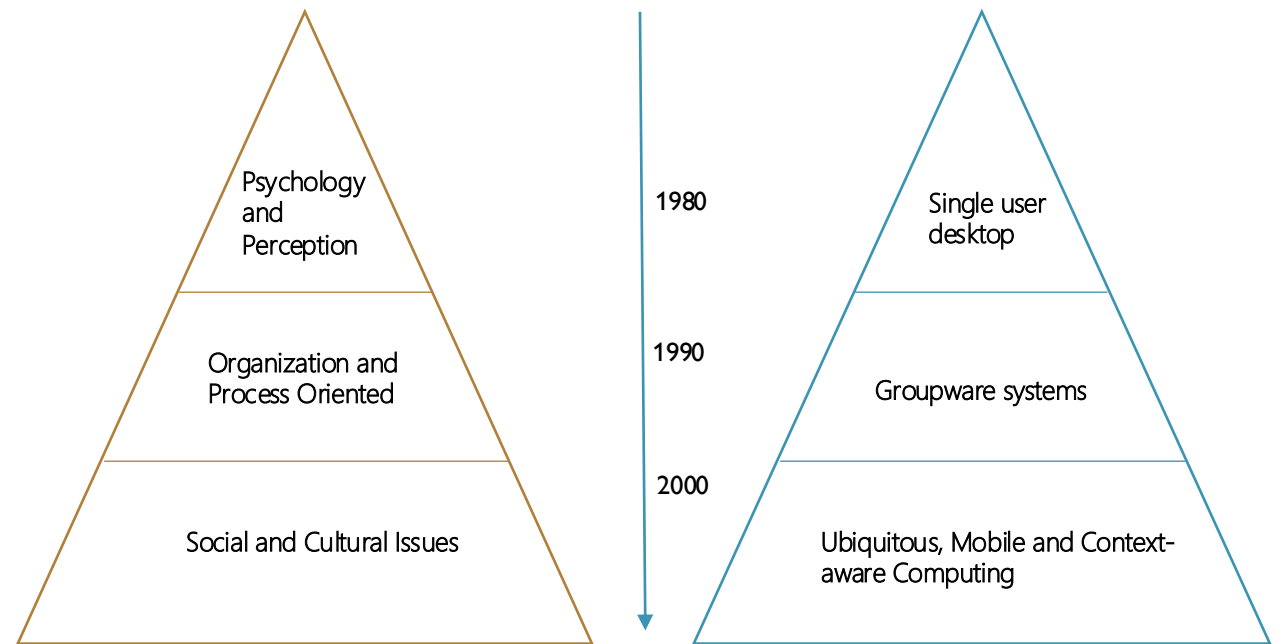
BROADENING METHODS

- Technology pushed progress here too
 - Eye-tracking studies, EEG...
- From early lab based studies that timed tasks/counted errors
- Brought in techniques more from sociology than psychology
 - Ethnography
 - Interviews
 - Case studies



HCI HAS PROGRESSED IN THREE “WAVES”

- First Wave: Psychology and Perception
- Second Wave: Organisational and Process Oriented
- Third Wave: Social and Ubiquitous
- Fourth Wave?



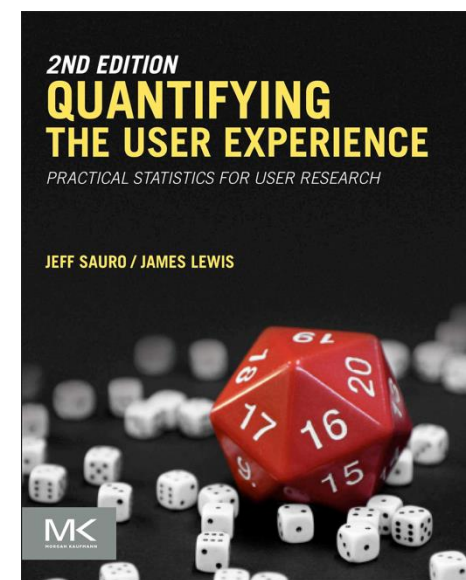
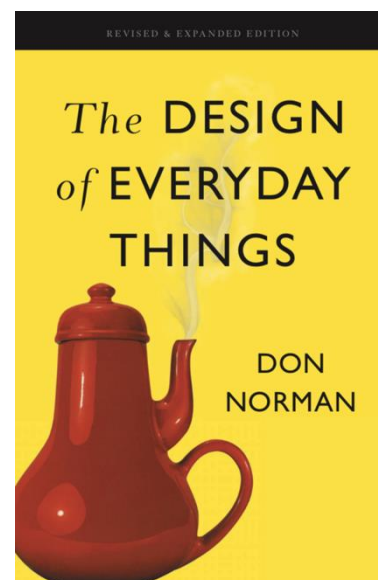
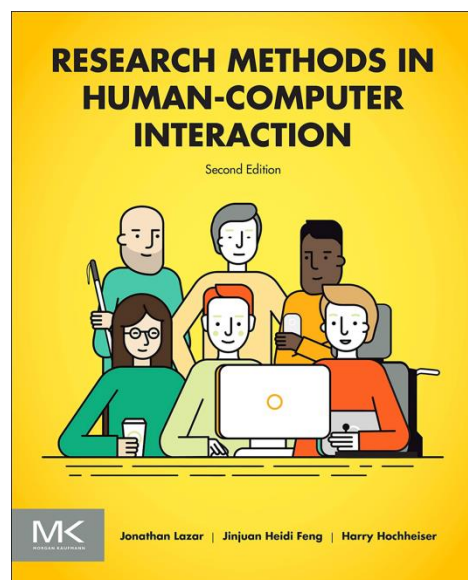
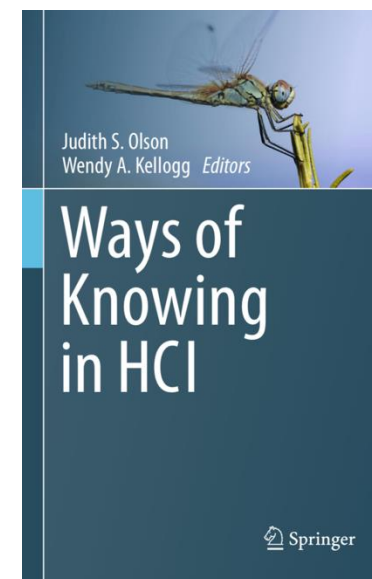
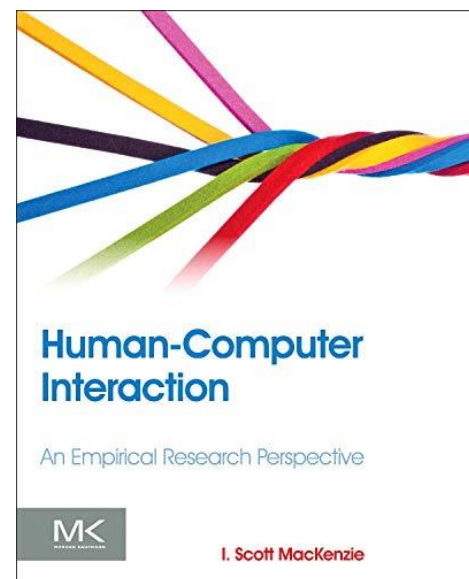
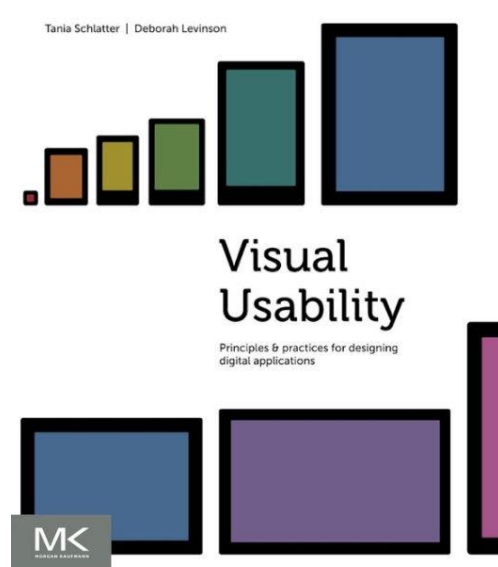


WHERE ARE WE NOW? WHAT IS THE
NEW PROGRESS, AND HOW HCI SHALL
RESPOND?
GO TO [MENTI.COM](https://www.menti.com) AND USE THE CODE

THE CLASS

HCIDE(M) TOPICS

- Visual usability
- Theories and heuristic evaluation
- Experiment design
- Quantitative analysis
- Qualitative methods
- Survey and mixed methods
- Creative design



ASSESSMENTS

Deadlines

- Group coursework
 - AE1: Heuristic usability evaluation (**formative**)
 - AE2: Experiment-based usability evaluation (**10%**)
 - AE3: Creative design (**10%**)
- Individual component
 - Participating in discussion in pre-AE3 (**5%**)
 - Final exam (**75%**)

5 Feb

19 Feb

19 Mar

April/May

- Group coursework
 - TAE1: Heuristic usability evaluation (15%)
 - TAE2: Experiment-based usability evaluation (25%)
 - TAE3: User-centred design (25%)
- Individual component
 - Qualitative coding (10%)
 - Online quiz (25%)

The year before last

- Group coursework
 - AE1: Heuristic usability evaluation (**5%**)
 - AE2: Experiment-based usability evaluation (**10%**)
 - AE3: Creative design (**10%**)
- Individual component
 - Final exam (**75%**)

Last year

COURSEWORK

- There will be a lab sheet on Moodle for each stage with more detailed explanation
- For group coursework
 - For summative coursework, each group receives a grade and textual feedback. Individual grades will be calculated based on peer assessment of each member's contribution
 - For formative coursework, general class feedback during lectures
- Individual participation component is based on your activeness and value in contributing to discussions

WHAT IS EXAMINABLE?

- All assigned readings as listed in Moodle
- All text of lecture slides

What is not examinable?

- Links provided in lecture slides labelled as “For Reference”

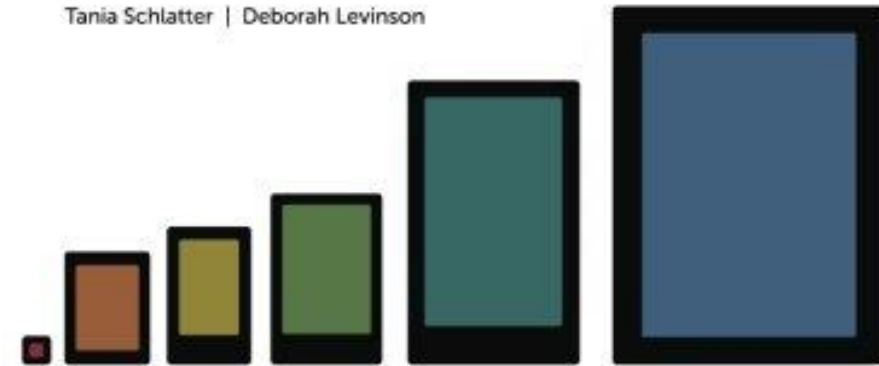
VISUAL USABILITY

*“Applying visual design
principles to complex
application design”*

Visual communication of any kind, whether
persuasive or informative ... should be seen
as the embodiment of form and function: the
integration of the beautiful and the useful.”

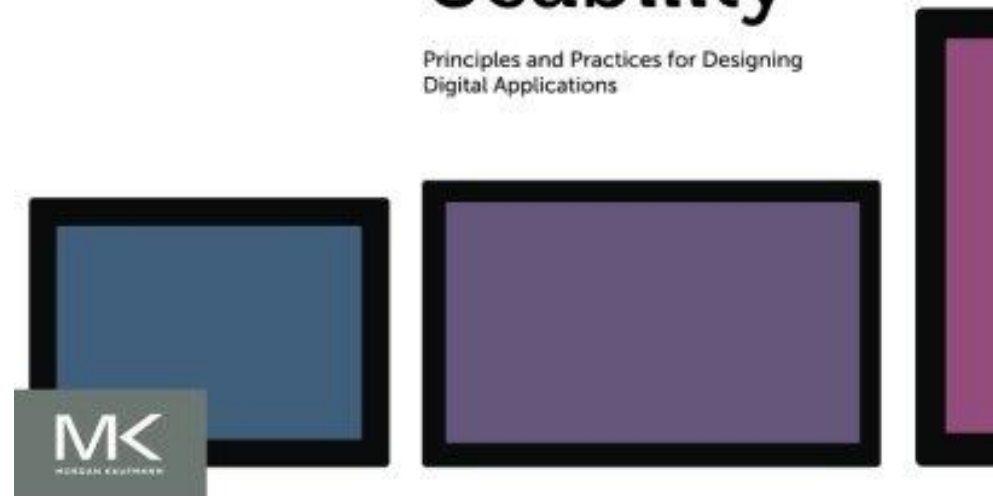
—Paul Rand, *A Designer’s Art*, p. 3

Tania Schlatter | Deborah Levinson



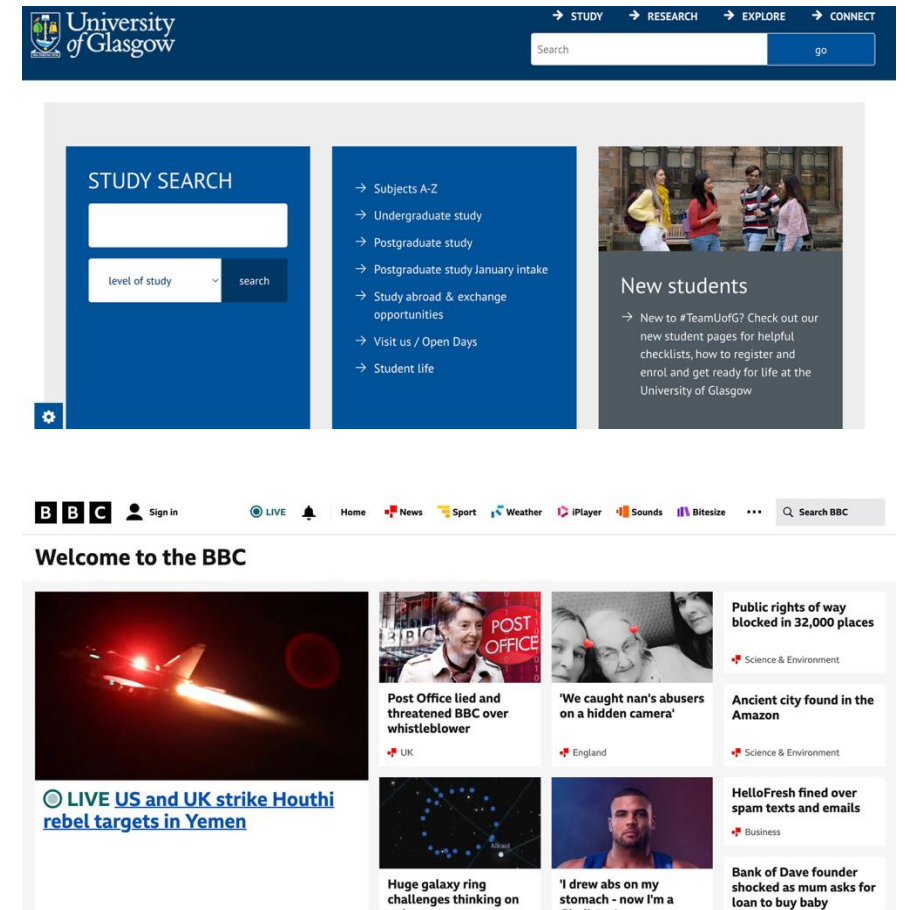
Visual Usability

Principles and Practices for Designing
Digital Applications



THE THINGS YOU SEE AROUND YOU TODAY ARE NOT THERE BY RANDOM CHANCE

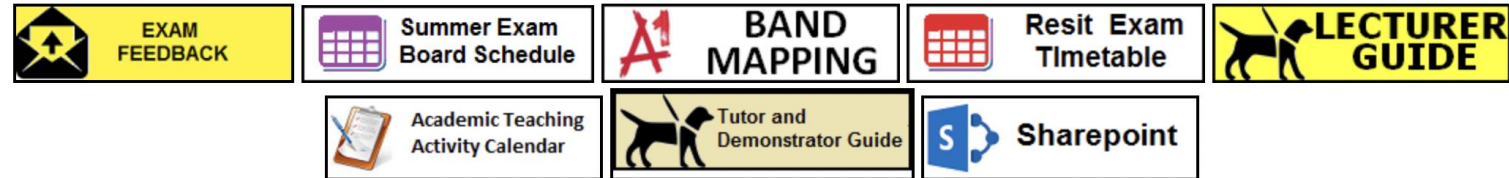
The interfaces familiar with us may seem easy to design, but are the result of many attempts and many failed designs.



CASE 1



Current Links For Staff.



{\LaTeX}
Latex Template



Word Template - Timed
Exam with Fixed Start Time



Word Template



Cover Sheet (Word)



Cover Sheet (PDF)



#name#
#name#

Where to find Soft Copies of Projects:

Level 4 Projects available from \\unix6\lev4proj\
Level 3 Projects available from \\unix6\lev3proj\
MRes Projects available from \\unix6\mscproj
MSci Projects available from \\unix6\lev5proj\
MSc Projects available from \\unix6\mscproj

Band Descriptors:

Band_Descriptors_Level3_Software_Engineering_Presentation
Band_Descriptors_Level3_Presentation.pdf
Band_Descriptors_Level3_Project.pdf
Band_Descriptors_Level4_Project.pdf

CASE 2

UofG Sport

[Logout](#)

Xianghua Ding

Welcome

Make a booking...

Book a sports activity

Book a class

Book a course

Contact Us

Your Bookings

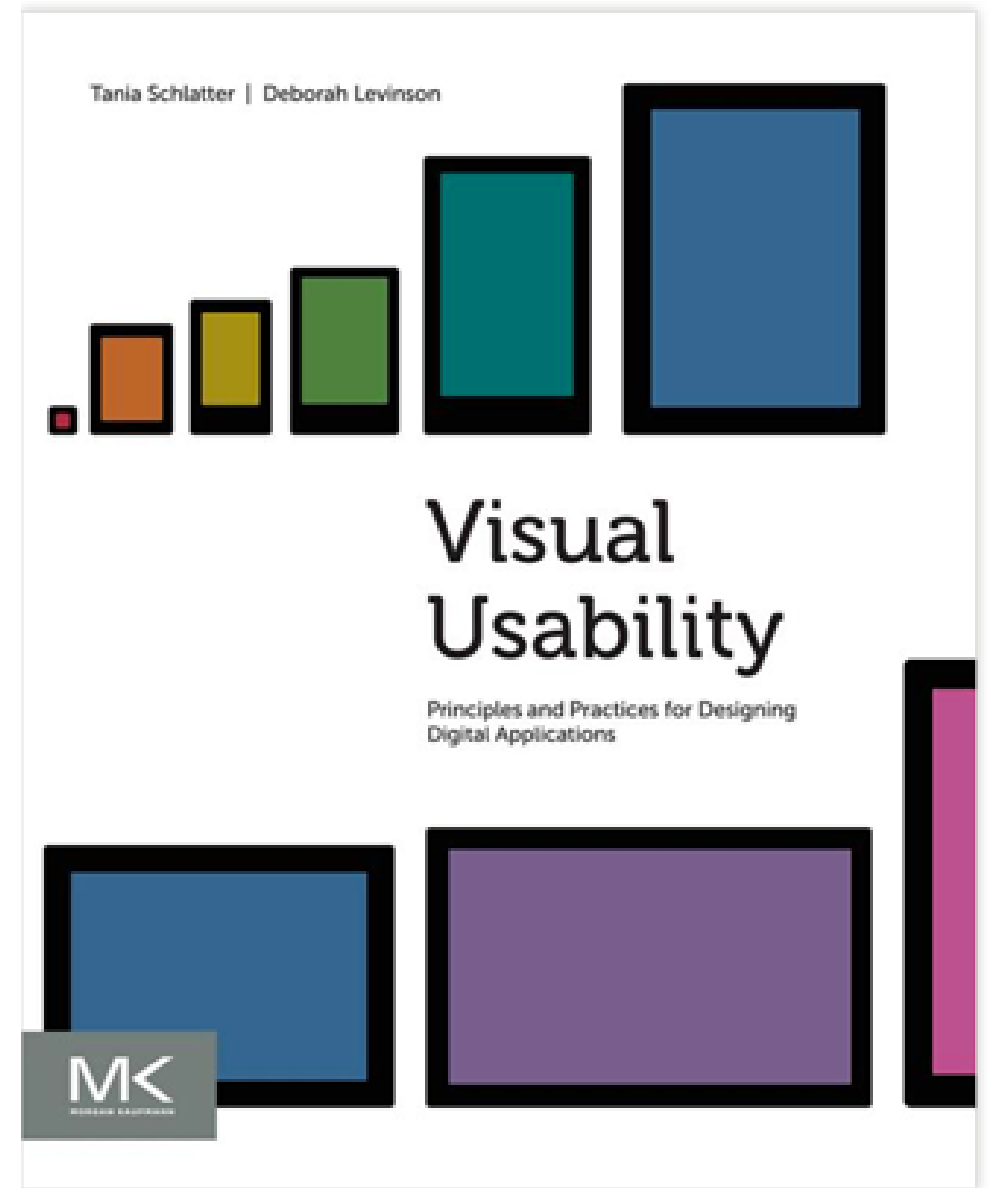
Your Courses

Contact Preferences

**Join or Top-Up
here**

VISUAL USABILITY

- Two meta-principles
 - Consistency
 - Hierarchy



CONSISTENCY



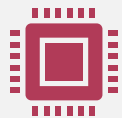
Establishing consistency means setting and maintaining expectations



External Consistency

Consistency with other similar applications

If you are designing an interface for online shopping, it should be similar to the established look/feel of existing interfaces



Internal Consistency

Consistency within different parts of an application

If you are designing an interface for online banking, all the views need consistency

ELEMENTS OF CONSISTENCY

Layout Consistency

- Screens showing similar information should have all elements positioned the same way
- Spatial relationships should remain consistent

Typography Consistency

- Use fonts, weights, and sizes meaningfully and consistently

Colour Consistency

- Use colour to establish and maintain consistency - typically means establishing a defined colour scheme

Imagery Consistency

- Charts, logos, videos, photography, icons, backgrounds, and anything else that isn't typography

Control and affordances

- Use the same interface elements and design treatments for the same actions
- Use the same motions for feedback and interactive controls

HIERARCHY

Visual hierarchy is used to communicate structure, relationships, and relative importance



More important items need more “visual weight”

Use position, size, colour, groupings, contrast, control types to represent priorities

THE LANGUAGE OF LAYOUT

Screen size

- The screen gives the frame within which your entire interface sits
- Core layout principles might apply to all screen sizes – main thing is how elements relate to each other, so layout can flex

Position

- Does the relative position of elements communicate structure
- Might need to balance lots of relationships

White Space

- Absence of content (not necessarily 'white'!) is equally important, for example the sparse design on a Google landing page
- Trick to create dense but appealing screens is white space to group and establish hierarchy
- Margin

Proximity, Scale and Alignment

- Is the relative placement and size and of items arbitrary or meaningful?
- Does the alignment to support grouping

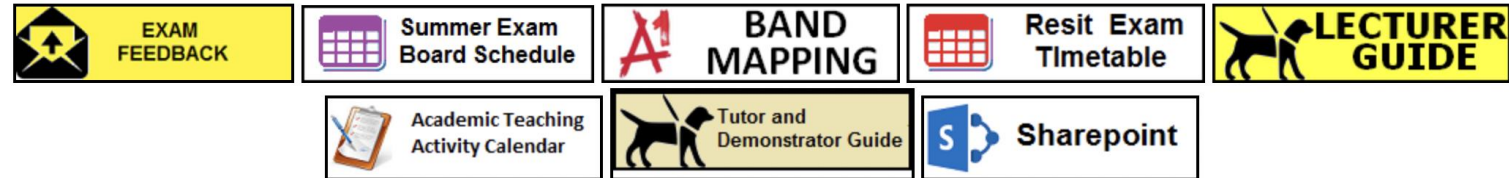
Grid

- Align items relative to (invisible) horizontal and vertical lines

CASE 1



Current Links For Staff.



Exams



Coursework



Projects



{\LaTeX}
Latex Template



Word Template - Timed
Exam with Fixed Start Time



Word Template



 University - School of
Glasgow - Computing Science
Business Computing

Cover Sheet (Word)

 University - School of
Glasgow - Computing Science
Business Computing

Cover Sheet (PDF)



#name#
#name#

Where to find Soft Copies of Projects:

Level 4 Projects available from \\unix6\lev4proj\
Level 3 Projects available from \\unix6\lev3proj\
MRes Projects available from \\unix6\mscproj
MSci Projects available from \\unix6\lev5proj\
MSc Projects available from \\unix6\mscproj

Band Descriptors:

Band_Descriptors_Level3_Software_Engineering_Presentation
Band_Descriptors_Level3_Presentation.pdf
Band_Descriptors_Level3_Project.pdf
Band_Descriptors_Level4_Project.pdf

CASE 2

UofG Sport

[Logout](#)

Xianghua Ding

Welcome

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**Join or Top-Up
here**

COLOR



COLOUR

- Powerful way of attracting the eye
 - Can create emotional response
 - Enhance usability and appeal
 - Aid understanding by creating connections between related items
- Colour choices
 - Can be cultural associations
 - Specific UI conventions
 - e.g. Red for error messages. But maybe only if critical
- Shouldn't convey anything crucial through colour alone
 - Visually impaired/colour-blind users